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Roll No

EC-7004(1)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Data Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Make suitable assumptions wherever necessary.

1. a) Explain the Circuit switching and Packet switching with the help of suitable timing diagrams. Compare and contrast the two on the basis of the following attributes: 8
 - i) Node delay
 - ii) Node failure
 - iii) Transmission time
 - iv) Propagation delay
- b) Draw the OSI model. Explain the functions of each layer in brief. 6
2. a) Describe the Asynchronous transmission of method for synchronization using the required character format? What do you mean by framing error? Explain with the help of suitable figure. 8
- b) What are the different services provided by ISDN? Explain in detail. 6

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3. a) Describe the sliding window flow control mechanism using suitable timing diagrams. 7
b) What is the maximum window size in selective repeat ARQ? Justify your answer with an example. 4
c) Why would you expect a CRC to detect more errors than a parity bit? Justify your answer. 3
4. a) Explain the three persistence protocols that can be used with CSMA? 6
b) Explain the architecture of Wireless LAN 802.11 and services offered by it. 8
5. a) What is the difference between a Datagram and Virtual circuit operation? 5
b) Explain the purpose and function of the ATM Adaptation Layer type 5 (AAL5) protocol. 9
6. a) By considering the operation of the TCP protocol, explain how it is able to overcome errors in the transmission and ensure that data is transferred reliably over a network. 7
b) Compare and contrast the IPv4 and the IPv6 header fields. Do they have any fields in common? 5
c) What is the 32-bit binary equivalent of the IP address 223.1.3.27? 2
7. a) Explain semantics of various fields in DNS message. 5
b) Describe the architecture of SNMP in detail. 9
8. Write short notes on any two of the following: 2×7=14
 - i) X.25
 - ii) HDLC
 - iii) BISDN

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